IIA WEB DESIGN • UNIT 11

Web Technologies in Information Institutions

Catalogues • digital repositories • archives • records systems • content management

Bachelor of Library and Information Studies | Bachelor of Records Management Systems

Learning outcomes

Recognise institutional systems

Distinguish an information webpage, a web application, a catalogue, a repository and a managed records service.

Connect concepts to practice

Identify system needs, data and metadata, permissions, workflow and the roles of information professionals.

KEY IDEA The focus is recognising and applying web technologies, not programming an entire institutional system.

From webpage to information service

Information webpage

A library may publish hours, service rules, staff contacts and reading guides using ordinary webpages.

Information service

A catalogue can search managed data and return book details; a records portal can support approved requests. These need more than static content.

KEY IDEA The visible webpage may be only one layer of a larger information system.

What a visitor sees is the frontend

Frontend

HTML defines forms and results; CSS styles navigation and layouts; JavaScript may help with immediate interactions.

Behind the interface

An application may connect the visitor to data and authorised processes while deciding what the visitor is allowed to see.

KEY IDEA A polished search box does not automatically mean a working catalogue exists.

Browser, backend and data: three layers

Browser layer

A person enters a query and reads results. The interface should be understandable, accessible and mobile friendly.

Service and data layers

A backend handles requests and rules. A database or managed file store supplies matching information.

KEY IDEA The browser asks; the application checks and retrieves; the interface shows a suitable response.

Example: find a library book

Visitor action

A student searches by title or author in an online library catalogue.

System response

The application searches catalogue records and displays matching titles, call numbers or locations; availability may also be displayed.

KEY IDEA A meaningful result depends on accurate underlying collection records.

What is an online public access catalogue?

OPAC: discovery interface

An OPAC lets visitors search for library materials using descriptive records such as title, author, subject and identifier.

Typical output

A record may show edition, publication details, location and item availability, depending on how the system is configured.

KEY IDEA The catalogue is a discovery tool, not merely an HTML table of sample titles.

Integrated library system: wider than the OPAC

ILS purpose

An ILS supports related operations such as cataloguing, circulation, acquisitions and patron services.

Public and staff views

Visitors may search a public catalogue; authorised staff may update records or manage library processes.

KEY IDEA An OPAC is often part of the larger integrated library system.

Koha: an ILS example

System type

Koha is a free and open-source integrated library system used by libraries of different types.

Professional relevance

Librarians work with data quality, cataloguing, service policies, search and user needs. Technical teams help configure and maintain the system.

KEY IDEA Recognise the type of system and its purpose before discussing software brands.

The value of bibliographic metadata

Descriptive fields

Title, creator, subject, publication information and call number help people identify and retrieve materials.

If data is poor

Misspelled titles or inconsistent subject terms can make a correct book difficult to discover even when the website looks attractive.

KEY IDEA Good information services depend on well-managed metadata, not visual design alone.

What is an institutional repository?

Repository role

A digital repository stores, describes, preserves and gives access to digital resources such as theses, articles, datasets or images.

Visitor journey

Search or browse → open an item record → examine metadata → access an approved digital file when available.

KEY IDEA A repository provides digital-content management and discovery rather than ordinary book circulation.

DSpace: a digital repository example

DSpace purpose

DSpace is an open-source platform for institutional and other digital repositories. It supports managed access to digital scholarship.

Example use

A university may deposit approved dissertations with a title, author, date, abstract, subject terms and a downloadable file.

KEY IDEA Different policies may apply to openly available, embargoed or restricted resources.

Catalogue and repository: compare

Catalogue focus

Often describes books and other collection items; a physical item may remain on a shelf.

Repository focus

Often manages the digital object itself as well as its metadata, discovery and permitted access.

KEY IDEA Both support discovery, but their materials and workflows may differ.

What makes archives distinctive?

Context and relationships

Archival records make sense through provenance, creation context, original arrangement and relationships across files and series.

Discovery need

A researcher may need a description of a collection and its reference code before requesting access to items.

KEY IDEA Archival discovery often depends on collection context, not only individual keywords.

Finding aids in archival services

A finding aid

A structured description guides researchers to collections, series, folders or related material.

Not the same as the item

Finding an archival description does not guarantee that the original file has been digitised or can be downloaded immediately.

KEY IDEA A finding aid supports access and discovery without necessarily storing every original object.

ArchivesSpace: an archives application

Primary role

ArchivesSpace supports archives information management, descriptions and public web access to holdings.

Important distinction

The platform is not a complete digital asset-management or digitisation workflow system; digital objects may be stored elsewhere.

KEY IDEA Choose software by the function it actually supports.

Records are more than electronic files

Document versus record

A document is information recorded in a form; a record is retained as evidence of an activity or transaction according to organisational requirements.

Management needs

Records may need identifiers, context, access controls, retention rules and evidence of actions taken.

KEY IDEA Moving a PDF to a public webpage does not create a compliant records-management process.

ERMS and EDRMS explained

ERMS

An electronic records-management system helps capture, classify, maintain, retrieve and dispose of records under controlled rules.

EDRMS

Some systems combine document-working features with formal records-management controls; capabilities vary between products.

KEY IDEA The central idea is managed records, not simply a folder full of digital documents.

Records lifecycle: capture and classify

Capture or receipt

Identify the record and register it into an organised system with a reference and essential context.

Classify

Associate the record with a file plan, category, responsible unit and appropriate access rules.

KEY IDEA Capture and classification support reliable identification and future retrieval.

Records lifecycle: use and disposition

Maintain and use

Keep information findable and protected while staff carry out authorised work.

Disposition

Retain, transfer or dispose of records under approved policies and applicable retention schedules.

KEY IDEA Deletion is not arbitrary; disposition is governed by established rules.

Why do audit trails matter?

Accountability

An audit trail may record meaningful actions such as changes, access or disposal events.

Reliability

Together with access controls and metadata, these records can help preserve integrity and explain what happened.

KEY IDEA An ordinary HTML table alone does not provide secure audit history.

What is a content management system?

CMS function

A CMS helps authorised staff create, edit, organise and publish website pages, announcements and information guides.

Professional use

A college library may need several staff to update public notices without manually editing HTML files.

KEY IDEA A CMS chiefly supports website content; it does not automatically replace an ILS or ERMS.

WordPress and Drupal as CMS examples

WordPress

Commonly used for websites, news and content publication where administrators update material through an interface.

Drupal

A flexible CMS that can organise complex website content and user roles according to configuration.

KEY IDEA CMS choice depends on users, security needs, content complexity and available support.

One institution, multiple systems

Library journey

A college homepage links to the OPAC for books and the repository for approved theses. Each system has a distinct role.

Records journey

A public guidance page explains requests; a secure records system handles controlled internal workflows.

KEY IDEA Clear links and consistent wording help users move between connected services.

What is an API?

Meaning

An application programming interface defines how software components can request or exchange information.

Catalogue example

A web interface may ask an application service for search results and then display them in the browser.

KEY IDEA Users do not need to see an API to benefit from a connected service.

Databases and search results

Stored information

An institutional database may hold descriptive records, item status or authorised workflow data.

Search process

The application performs queries and returns selected information according to its business and permission rules.

KEY IDEA A database is not the same as an HTML table shown in a browser.

Authentication versus authorisation

Authentication

Checks who the user is, for example through an institutional sign-in.

Authorisation

Determines which resources and actions that user is permitted to access after identity is established.

KEY IDEA A public visitor and a registered staff member may have different rights in one system.

Public information and restricted records

Suitable public content

Opening hours, approved catalogues, public service guidance and permitted research outputs.

Sensitive content

Borrower data, confidential personnel records or restricted office documents require appropriate protection.

KEY IDEA Professional judgement about access is as important as interface design.

Why a hidden link is not security

Visual hiding

CSS can hide a control from sight, but this does not enforce a server-side rule on who may retrieve the underlying resource.

Real protection

A trusted service needs appropriate authentication, authorisation, validated requests and protected storage.

KEY IDEA Interface behaviour must never be confused with access control.

Metadata, integrity and preservation

Metadata

Describes a resource and supports reliable identification, context, retrieval and management.

Integrity and preservation

Controls help prevent improper modification and preserve accessibility, with backups and additional long-term measures where necessary.

KEY IDEA Information professionals help protect trust in managed collections and records.

Which system meets the need?

Library situation

Students need to search titles and shelf locations. A catalogue and its supporting ILS are appropriate.

Records situation

Staff must apply retention rules and manage restricted files. A controlled records system is appropriate.

KEY IDEA Choose the system from the user task, the information type and access requirements.

Example: public enquiry website

Visitor need

A citizen wants opening hours, office location and clear public-records-request instructions.

Possible solution

A well-designed information website, perhaps updated through a CMS, may be sufficient for general public guidance.

KEY IDEA Do not make a system more complex than the task requires.

Guided practice: map the journey

Step 1 and 2

Choose a fictional visitor and define the task. Identify the website or application where the visitor begins.

Step 3 and 4

Decide which system stores the relevant information and identify one metadata or privacy requirement.

KEY IDEA A short arrow diagram can show how a real user reaches information.

Assignment 11: recommend a service

Choose a scenario

Select a library discovery and thesis-access need, or a public records guidance and restricted records need.

Prepare a brief response

In about 120–150 words, explain two useful features and one privacy or access consideration.

KEY IDEA The task checks understanding; no programming or installation is needed.

Review key concepts

Name the systems

OPAC/ILS, digital repository, archives description, ERMS/EDRMS and CMS serve different institutional needs.

Name the safeguards

Metadata, access rules, authentication, authorisation, audit history and preservation help keep services reliable.

KEY IDEA Connect technical features to the needs of information users and institutions.

Unit 11: essential takeaways

A website is an interface

The visible page may present public content or connect to specialised databases and workflows.

Information professionals matter

They define user needs, organise collections, protect records, improve metadata and assess appropriate access.

KEY IDEA Next: Learn how websites are hosted, published and maintained.